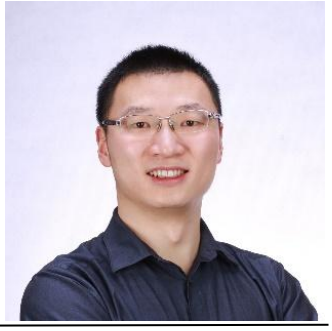
王者



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个人简介

王者，香港科技大学副教授，智慧城市研究院副院长，**2021 海外优青**，**2023 港澳优青**，曾任劳伦斯伯克利国家实验室科学家。清华大学本科（双学位，均排名第一），剑桥大学硕士，清华大学博士（导师：朱颖心教授、林波荣教授），美国加州伯克利大学博士后；北京市优秀毕业生（2011，2017）；清华大学首批苏世民学者（2016）、紫荆学者（2017），受邀以**中方唯一青年学者代表**身份在刘延东副总理、克里国务卿主持的第七届中美人文交流高层对话（2016）上发言，曾被纽约时报（New York Times）报道（2016 年 1 月 11 日）。

王者的主要研究方向是未来建筑节能低碳、智慧运维、智能热管理技术。面向我国新时期城镇化进程中的节能低碳、智能健康需求开展基础原理和创新技术研究，在(1)空间环境状态感知与大数据分析；(2)能源与环境系统智能控制优化算法；(3)数据中心与机器人热管理等三方面取得系列创新成果。2021 年至今，连续四年入选斯坦福大学和爱斯维尔联合发布的“全球前 2%顶尖科学家”榜单。王者现任 SCI 期刊 Advances in Applied Energy, Energy and Buildings, Building Simulation 编委；曾获 **2019 国家科技进步二等奖**，**2018 北京市科学技术一等奖**，2020 华夏建设科学技术一等奖；获香港机电署举办的 2022 国际建筑机电人工智能挑战赛第一名，NeurIPS CityLearn Challenge 2023 第二名。

工作经历

2021.11 至今 **香港科技大学，助理教授(21-25)，副教授(25-)，智慧城市研究院副院长(25-)，香港太空机器人与能源中心管理委员会成员（26-）**

- PI，嫦娥八号月面多功能移动操作机械人暨移动充电站 RP11-热控分系统（17**00 万港币**，創新香港研發平台），2025-2029
- PI，“建筑群-电动汽车-电网”互动的柔性潜力挖掘及调控技术 (**200 万人民币**，2023 年度国家重点研发计划“城镇可持续发展关键技术与装备”重点专项)，2024-2027
- PI, 绿色智慧建筑热环境 (**200 万人民币**，国家自然科学基金委优秀青年基金（港澳），52322813)，2024-2026
- PI, 基于动态热舒适的电动汽车座舱热环境管理与多目标优化研究 (**30 万人民币**，国家自然科学基金委青年基金，52306028)，2024-2026
- PI, Semantic AI empowered fault detection and control optimization for HVAC system of large-scale commercial buildings (**90 万港币**, the Innovation and Technology Fund by ITC, PRP-061-23FX), 2025-2028
- PI，Towards a human-centric smart cabin: System design integrating thermal comfort and air quality (**50 万港币**，HKUST – HKUST(GZ) 20 for 20 Cross-campus Collaborative Research Scheme，C006)，2024-2026
- PI, 基于无人机、机器视觉和多源数据融合的建筑结构探伤 (**100 万人民币**，深圳瑞捷工程咨询股份有限公司研发课题)，2024-2025
- PI, 增强数据中心的运行效率与可靠性 (**20 万人民币**，2024 腾讯基础平台技术犀牛鸟专项研究计划)，2024-2025
- PI, 多联机系统需求响应算法研究 (**60 万人民币**，美的研发课题)，2024
- PC，Toward 2060 Carbon Neutrality: Life-cycle Planning and Design of Photovoltaic Integrated Green Roof (PVIGR) Systems for Hong Kong and the Greater Bay Area (**346 万港币**, 香港研究资助局合作研究项目，C6003-22Y)，2023-2026
- PI，Model Predictive Control in Hong Kong’s Residential Buildings for Energy Efficiency and Load

Flexibility (**84 万港币**, 香港研究资助局杰出青年学者计划，26209323)，2023-2026

- PI，公共建筑空调系统能效诊断与优化控制研究 (**80 万人民币**，深圳市科委可持续发展科技专项 KCXST20221021111403009)，2023-2026
- PI，An AI-assisted solution for low-cost high-resolution urban scale environmental simulation (**36 万港币**, Fei Chi En Education and Research Fund)，2023-2025
- PI，建筑能源系统模拟与优化研究 (**30 万人民币**，瑞安绿色可持续发展研究基金)，2023-2025
- PI，室内消毒与环境控制智能机器人 (**72 万人民币**, 佛山市政府课题 SHCIRI-FSNH-2203)，2022-2024
- PI, 电力需求响应软件架构研究 (**60 万人民币**，美的研发课题)，2022-2023
- PI, 超大型数据中心制冷系统智能化节能方案研究 (**30 万人民币**，2022 腾讯基础平台技术犀牛鸟专项研究计划)，2022-2023

2018.07-2021.10 **美国劳伦斯伯克利国家实验室，项目科学家**

- co-PI，Reinforcement Learning for Building Control (**0.2M USD**, 美国 LBNL 课题: BU21-036)
- Integrating Sensor Data with Physics-Based Models (**1.5M USD**, 美国能源部课题: EE-5B37579)
- Hierarchical Occupancy Responsive MPC (**3.0M USD**, 美国能源部课题: EE-5B24502)

2017.10-2018.06 **加州伯克利大学，博士后研究员**

- Personalized Wearable Comfort Devices (**0.5M USD**，美国 NSF 课题)

2016.12-2018.06 **世界银行，能源咨询专家**

- 世界银行&住建部：中国城市建筑节能和可再生能源应用项目，咨询专家

教育经历

2011.08-2017.07 **清华大学，建筑技术，博士**

- 哈佛大学访问学者，2015
- 获得：国家奖学金 (2012)，清华大学综合优秀奖学金 (2012, 2015)，波音奖学金 (2015)，北京建筑设计院优秀学生奖学金 (2016)，北京市优秀毕业生 (2017)，建筑学院优秀博士论文 (2017)
- 共青团清华大学建筑学院委员会书记，2011-2013

2016.08-2017.07 **清华大学苏世民书院，公共政策，硕士**

- 受邀在 2016 年第七轮中美人文交流高层磋商全体会议（由中国副总理刘延东和美国国务卿克里主持）上发言，唯一中国青年学者代表
- 学生自治委员会成员 (110 名学生中选举产生 12 名)

2013.10-2014.09 **剑桥大学，能源技术，硕士**

- 获得：Wing Yip 奖学金 (每年 3 人)

2008.09-2011.07 **清华大学，经济学，学士**

- 平均绩点 92.8/100, 排名 **1** / 126

2007.09-2011.07 **清华大学，土木工程，学士**

- 平均绩点 91.3/100, 排名 **1** / 29
- 获得：国家奖学金 (2009, 2010)，波音奖学金 (2011)，北京市优秀毕业生 (2011)，清华大学优秀毕业生 (2011)，优秀学位论文 (2011)
- 清华大学学生网络电视台台长，2010-2011

开源软件

MPCPy Python-based open-source platform for model predictive control in buildings
https://github.com/lbl-srg/MPCPy

Modelica Buildings Library Dynamic simulation models for building energy and control systems
https://github.com/lbl-srg/modelica-buildings

学术服务

期刊编委	*Energy and Buildings,* **Section Editor** *in Intelligent Building and Smart Control*
	Building Simulation, **Subject Editor** in *Building Control*
	Advances in Applied Energy, **Assistant Editor**
学校行政	Associate Director of HKUST Smart City Institute
	Member of the University Committee on Student Affairs
基金评委	*Natural Sciences and Engineering Research Council Grant*, Canada, 2025
	Frontier Competitive Research Program, Singapore, 2024
	MTR Research Funding Scheme, HK, 2023
	The Dunhill Medical Trust, UK, 2020
会议组织	*Asian University Alliance Academic Conference on smart building*, **Conference Chair**, 2024
	Jiangsu-Hong Kong-Macau University Alliance Seminar on smart city and green building, **Conference Chair**, 2023
	International Workshop on Reinforcement Learning for Energy Management in Buildings & Cities (RLEM), https://rlem-workshop.net/, **Member of Technical Program Committee**, 2020, 2021, 2023
	The 13th *International Symposium on Heating, Ventilation and Air-conditioning (ISHVAC)*, http://ishvac2023.org/, **Member of Scientific Committee**, 2023
期刊审稿人	*Nature Energy, Applied Energy, Energy, Building and Environment, Energy and Building, Journal of Building Engineering, Building Simulation, Environmental Science and Pollution Research, Science and Technology for the Built Environment, SoftwareX, Journal of Building Performance Simulation, Applied Thermal Engineering, Advanced Engineering Informatics, Frontiers in Built Environment, International Journal of Refrigeration, International Journal of Biometeorology, Journal of Asian Architecture and Building Engineering, Journal of the Taiwan Institute of Chemical Engineer, Sustainable Cities and Society, Engineering*

获奖

1. 《绿色公共建筑环境与节能设计关键技术研究与应用》，2019 国家科学技术进步二等奖，排名第 9，证书号：2019-J-22101-2-01-R09
2. 2022 国际建筑机电人工智能挑战赛，第一名，香港机电署，学生第一作者，本人通讯作者（奖金 20 万港币）
3. 2023 NeurIPS CityLearn Challenge，第二名，学生第一作者，本人通讯作者
4. 《绿色公共建筑环境与节能设计关键技术研究与应用》，2018 北京市科学技术一等奖，排名第 10
5. 《公共建筑室内环境智能监控和节能关键技术研究》，2020 华夏建设科学技术一等奖，排名第 10
6. 《舒适性多联式空调系统全工况高精度运行节能技术及产业化》，2025 中国节能协会节能减排科技进步奖特等奖，排名第 5

人才称号

1. 港澳优青，2023
2. 海外优青，2021
3. 乔治布什总统基金会 Fellow，2021-2024
4. 清华大学紫荆学者，2017
5. 清华大学苏世民学者，2016

专利

1. **王者**，一种能响应分时电价和需求响应信号的智能温控器和智能温控器的控制方法，中国专利申请号：202210667572.X，初审合格
2. **王者**，张世鸿，陈柳涛，吴至复，李磊，王朝亮，刘炜，城市尺度建筑能耗模拟方法、系统、存储介质和产品，

中国专利申请号：202410909228.6，初审合格

3. **王者**，李斯琪，李书浩，冷机运行序列寻优方法、系统、存储介质和产品，中国专利申请号：202410909230.3，初审合格
4. **王者**，周期，郑烨，实时全局污染物浓度移动监测的路径规划方法和系统，中国专利申请号：202410909229.0，初审合格
5. 林波荣，**王者**，李紫微，裴祖峰，刘凯，高能效住宅建筑空调系统，中国专利号: ZL 201310161191.5
6. 林波荣，赵海湉，**王者**, 一种用于洗碗机的热回收装置，中国专利号: ZL 201320328067.9

学术文章

谷歌学术引用: 10234; h-index: 52; i10-index: 102; ESI 高被引论文: 6

(#: 共同第一作者; *: 通讯作者)

First Author

1. **Wang, Z.*** and He, Y., 2023. AlphaHydrogen: A virtual platform for simulating and evaluating station-based regional hydrogen-electricity networks with distributed renewables, buildings, and fuel-cell vehicles. Energy Conversion and Management, 280, p.116802.
2. **Wang, Z.***, 2022. How frequent should we measure the indoor thermal environment. Building and Environment, 222, p.109464.
3. **Wang, Z.**, Chen, B., Li, H. and Hong, T., 2021. AlphaBuilding ResCommunity: A multi-agent virtual testbed for community-level load coordination. Advances in Applied Energy, 4, p.100061.
4. **Wang, Z.**, Hong, T. and Li, H., 2021. Informing the planning of rotating power outages in heat waves through data analytics of connected smart thermostats for residential buildings. Environmental Research Letters, 16(7), p.074003.
5. **Wang, Z.**, Hong, T., Li, H. and Piette, M.A., 2021. Predicting city-scale daily electricity consumption using data-driven models. Advances in Applied Energy, 2, p.100025.
6. **Wang, Z.** and Hong, T., 2020. Reinforcement Learning for Building Controls: The opportunities and challenges. Applied Energy, 269, p.115036. (highly cited paper)
7. **Wang, Z.**, Hong, T. and Piette, M.A., 2020. Building thermal load prediction through shallow machine learning and deep learning. Applied Energy, 263, p.114683. (highly cited paper)
8. **Wang, Z.** and Hong, T., 2020. Learning occupants' indoor comfort temperature through a Bayesian inference approach for office buildings in United States. Renewable and Sustainable Energy Reviews, 119, p.109593.
9. **Wang, Z.** and Hong, T., 2020. Generating realistic building electrical load profiles through the Generative Adversarial Network (GAN). Energy and Buildings, p.110299.
10. **Wang, Z.**, Hong, T., Piette, M.A. and Pritoni, M. 2019, Inferring occupant counts from Wi-Fi data in buildings through machine learning, Building and Environment, 158, pp. 281-294.
11. **Wang, Z.**, Parkinson, T., Li, P., Lin, B. and Hong, T., 2019. The Squeaky wheel: Machine learning for anomaly detection in subjective thermal comfort votes. Building and Environment, 151, pp.219-227.
12. **Wang, Z.**, Zhang, H., He, Y., Luo, M., Li, Z., Hong, T. and Lin, B., 2020. Revisiting individual and group differences in thermal comfort based on ASHRAE database. Energy and Buildings, 219, p.110017.
13. **Wang, Z.**, Wang, J., He, Y., Liu, Y., Lin, B. and Hong, T., 2020. Dimension analysis of subjective thermal comfort metrics based on ASHRAE Global Thermal Comfort Database using machine learning. Journal of Building Engineering, 29, p.101120.
14. **Wang, Z.**, Hong, T. and Piette, M.A., 2019. Predicting plug loads with occupant count data through a deep learning approach. Energy, 181, pp.29-42.
15. **Wang, Z.**, Warren, K., Luo, M., He, X., Zhang, H., Arens, E., Chen, W., He, Y., Hu, Y., Jin, L. and Liu, S., 2019. Evaluating the comfort of thermally dynamic wearable devices. Building and Environment, p.106443.
16. **Wang, Z.**, Hong, T. and Piette, M.A., 2019. Data fusion in predicting internal heat gains for office buildings through a deep learning approach. Applied Energy, 240, pp.386-398.
17. **Wang, Z.**, Hong, T. and Jia, R., 2018. Buildings. Occupants: a Modelica package for modelling occupant behaviour in buildings. Journal of Building Performance Simulation, pp.1-12.

18. **Wang, Z.**, Luo, M., Geng, Y., Lin, B. and Zhu, Y., 2018. A model to compare convective and radiant heating systems for intermittent space heating. Applied Energy, 215, pp.211-226.
19. **Wang, Z.**, de Dear, R., Luo, M., Lin, B., He, Y., Ghahramani, A. and Zhu, Y., 2018. Individual difference in thermal comfort: A literature review. Building and Environment, 138, pp. 181-193 (highly cited paper)
20. **Wang, Z.**, Zhao, Z., Lin, B., Zhu, Y. and Ouyang, Q., 2015. Residential heating energy consumption modeling through a bottom-up approach for China's Hot Summer–Cold Winter climatic region. Energy and Buildings, 109, pp.65-74.
21. **Wang, Z.**, Zhao, H., Lin, B., Zhu, Y., Ouyang, Q. and Yu, J., 2015. Investigation of indoor environment quality of Chinese large-hub airport terminal buildings through longitudinal field measurement and subjective survey. Building and Environment, 94, pp.593-605.
22. **Wang, Z.**, de Dear, R., Lin, B., Zhu, Y. and Ouyang, Q., 2015. Rational selection of heating temperature set points for China's hot summer–Cold winter climatic region. Building and Environment, 93, pp.63-70.
23. **Wang, Z.**, Lin, B. and Zhu, Y., 2015. Modeling and measurement study on an intermittent heating system of a residence in Cambridgeshire. Building and Environment, 92, pp.380-386.

Co-first Author

24. Liu, S.*,#, **Wang, Z.***,#, Schiavon, S., He, Y., Luo, M., Zhang, H. and Arens, E., 2020. Predicted percentage dissatisfied with vertical temperature gradient. Energy and Buildings, p.110085.
25. Wang, J.#, **Wang, Z.**#, Zhou, D. and Sun, K., 2019. Key issues and novel optimization approaches of industrial waste heat recovery in district heating systems. Energy, p.116005.
26. Wang, J.#, **Wang, Z.** *,#, de Dear, R., Luo, M., Ghahramani, A. and Lin, B., 2018. The uncertainty of subjective thermal comfort measurement. Energy and Buildings, 181, pp.38-49.
27. Liu, Y.#, **Wang, Z.**#, Lin, B., Hong, J. and Zhu, Y., 2018. Occupant satisfaction in Three-Star-certified office buildings based on comparative study using LEED and BREEAM. Building and Environment, 132, pp.1-10.
28. Touzani, S.#, Prakash, A.K.#, **Wang, Z.**#, Agarwal, S., Pritoni, M., Kiran, M., Brown, R. and Granderson, J., 2021. Controlling distributed energy resources via deep reinforcement learning for load flexibility and energy efficiency. Applied Energy, 304, p.117733.

Corresponding Author

29. Hu, Y., Lu, K., Duan, Z., Chen, H. and **Wang, Z.***, 2026. Cross-modal object detection for UAV-based multispectral delamination assessment of building envelopes. Automation in Construction, 184, p.106857.
30. Li, S., Li, S., Liu, Y., Li, B., Zheng, C. and **Wang, Z.***, 2026. Model predictive control of variable refrigerant flow systems for demand response: balancing the trade-off between model accuracy and optimization complexity: a case study of indoor system. Energy Conversion and Management, 357, p.121400.
31. Chen, L., Zhang, S., Chui, T.F.M., Yang, J., Wang, W., Hong, T. and **Wang, Z.***, 2026. Mitigating urban climate-energy feedback with citywide building-integrated photovoltaics implementation. Nexus, 3(1).
32. Li, L., Pu, Y., Ju, Y., Hu, Z. and **Wang, Z.***, 2026. Coordinating Multiple UPS Batteries in Datacenter for Load Flexibility and Cost Reduction. Energy and Buildings, p.117282.
33. Li, M., Hu, Z., Mohebi, P., Li, S. and **Wang, Z.***, 2026. Enhancing LLM-based building data query with chain-of-thought, retrieval-augmented generation, and fine-tuning. Automation in Construction, 182, p.106738.
34. Mohebi, P., Hu, Z., Li, L., Golzar, F. and **Wang, Z.***, 2026. Optimal battery sizing using stochastic programming to consider building load variation and peak demand charge. Energy Conversion and Management, 348, p.120794.
35. Zheng, W., Liu, L., Li, S., Zhang, Z., Zhao, P., Li, T., Zhang, X. and **Wang, Z.***, 2026. Smart predict-then-optimize-based model predictive control for Virtual Power Plants with battery storage. Journal of Energy Storage, 142, p.119606.
36. Huang, Q. and **Wang, Z.***, 2026. Towards energy-positive and eco-profitable future: A perspective on building-to-vehicle-to-building ecosystems. Sustainable Energy Technologies and Assessments, 85, p.104760.
37. Wang, Z., Sheng, H., Gu, F., Zhao, L., **Wang, Z.*** and He, D., 2026. The influence of cabin environment on takeover performance in conditional automated driving. Transportation Research Part F: Traffic Psychology and Behaviour, 117, p.103478.
38. Wang, Z., Sheng, H., Gu, F., Zhou, Y., Zhao, L., **Wang, Z.*** and He, D., 2025. Investigating effects of temperature and CO2 on driver drowsiness in the context of conditional automated driving. Ergonomics, pp.1-18.
39. Li, S., Li, M., Hu, Z. and **Wang, Z.***, 2025. Applying semantic model for easy and fast deployment of chiller sequencing

algorithm. Energy and Buildings, p.116830.
40. Li, M. and **Wang, Z.***, 2025. BuildingGPT: Query building semantic data using large language models and vector-graph retrieval-augmented generation. Building and Environment, p.113855.
41. Zhang, D., Li, M. and **Wang, Z.***, 2025. PV-GPT: query the PV data using natural language. Journal of Building Engineering, p.114535.
42. Hu, Z., Li, M., Tang, H. and **Wang, Z.***, 2025. AutoControl: An end-to-end fully automated workflow for control design of building energy systems. Energy, p.138329.
43. Duan, Z., Lu, K., Li, M., Zhang, S., Lin, B. and **Wang, Z.***, 2025. Automating exterior BIM for existing buildings using GIS data and UAV-based 3D modeling. Automation in Construction, 178, p.106422.
44. Zhao, L., Mohebi, P. and **Wang, Z.***, 2025. A Model-Based EV Driving Range Assessment with Genetic-Algorithm-Optimized Motor Waste Heat Recovery Strategies. Energy Use, 1(1), pp.100005-1.
45. Chen, L., Zhang, S., Cheng, I., Chang, H., Chen, F., Li, M. and **Wang, Z.***, 2025. The Resilience Paradox of Rooftop PV: Building Cooling Penalties and Heat Risks. Building and Environment, p.113233.
46. Wu, S., Yang, P., Li, D., Chen, G., and **Wang, Z.**, 2025. Towards Predictive Maintenance: A Performance Evaluation Framework for Cooling Towers in HVAC Systems. Building and Environment, p.113443.
47. Li, S., Li, S., Mohebi, P., Wang, D., Ma, M.N., Liu, G. and **Wang, Z.***, 2025. Field demonstration of model predictive control for chiller sequencing in large-scale commercial buildings. Energy and Buildings, p.116021.
48. Chen, L., Lin, Z., Zhou, Q., Zhang, S., Li, M. and **Wang, Z.***, 2025. Impacts of photovoltaics and integrated green roofs on urban climate: Experimental insights for urban land surface modelling. Renewable and Sustainable Energy Reviews, 217, p.115709.
49. Wang, D., Zheng, W., Li, S., Chen, Y., Lin, X. and **Wang, Z.***, 2025. Impact analysis of uncertainty in thermal resistor-capacitor models on model predictive control performance. Energy and Buildings, 328, p.115112.
50. Zheng, W., Hu, Z., Wang, D. and **Wang, Z.***, 2025. Optimizing building energy systems for grid-interactivity, comfort and resilience. Energy Conversion and Management, 340, p.119927.
51. Wu, S., Zheng, W., **Wang, Z.***, Chen, G., Yang, P., Yue, S., Li, D. and Wu, Y., 2025. AlphaDataCenterCooling: A virtual testbed for evaluating operational strategies in data center cooling plants. Applied Energy, 380, p.125100.
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